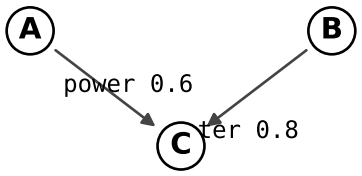
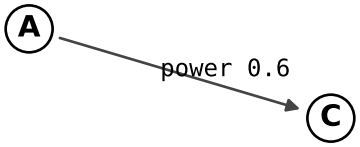
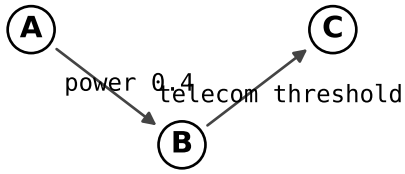
**Weakest supply wins**
DRAFT wording
A node works only as well as its weakest supply. Its state is the lowest of its responses to everything it needs (minimum across shortfalls, not their product).
**Registry**
[failure_modes.combining]



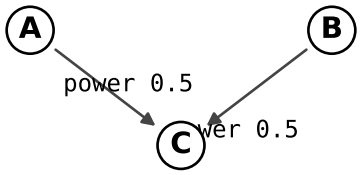

**Throttle, trip and floor**
DRAFT wording
Short of a supply, a node throttles (runs at the share it gets), trips (runs fully while it gets at least 0.9, else stops) or, on its generator, steps to a floor of 0.6.
**Registry**
= 0.9 ratio of supply received
[dynamics.trip_point]
= 0.6 ratio of normal load
[reserves.hospital .essential_load_fraction]



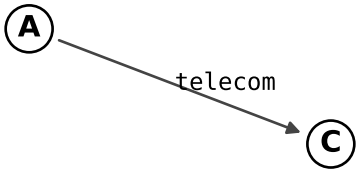

**How a partial supply passes**
DRAFT wording
An edge passes its supplier's state on in one of two ways. Linear: it passes κ × the state. Threshold: it passes all while the supplier is at least 0.5, else nothing.
**Registry**
= 0.5 ratio of supplier state
[dynamics.threshold]



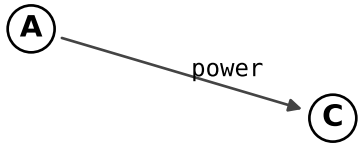

**Two supplies into one node**
DRAFT wording
Two feeds of one supply add up, each giving κ × its supplier's state, to at most 1. Within its own network, a node takes the best of its incoming links.



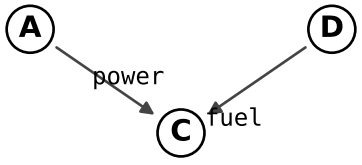

**Telecom fallback**
DRAFT wording
When telecom is lost, a node with a fallback keeps full function for T hours, then drops to d_eff = 1 − κ(1 − d). A clinic: T = 24 h, d = 0.8.
**Registry**
= t_h 24, d 0.8
[tolerance.clinic]



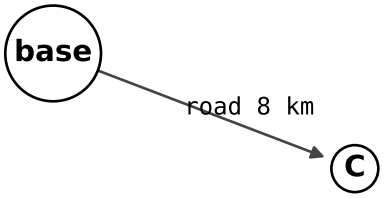

**Stocks drain and refill**
DRAFT wording
A stock covers its own supply while it lasts, at the node's draw. Batteries recharge empty to full in 12 h once power is back; water from the main refills at 1 × the drain rate.
**Registry**
= 12 h
[reserves .battery_recharge_h]
= 1 ratio of the drain rate
[reserves .main_refill_rate_fraction]



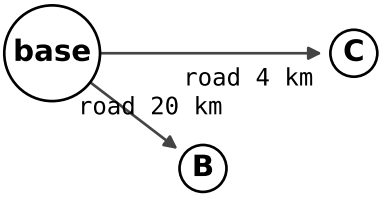

**Fuel deliveries**
DRAFT wording
A generator tank is refilled by delivery from a working depot: the first 12 h after power is lost, then every 24 h, each bringing 24 h of fuel.
**Registry**
= 12 h
[reserves.fuel_resupply .lead_time_h]
= 24 h of fuel
[reserves.fuel_resupply .delivery_h]




**The crew sequence**
DRAFT wording
A crew's job runs in order: travel, then 2 h to assess and prepare, then wait for any parts, then the work. Roads at 40 km/h.
**Registry**
= 2 h
[repair.crew_bound_lead_h]
= 40 km/h
[crews.road_speed_kmh]




**Crew priority order and ties**
DRAFT wording
A free crew takes the most urgent job first: towered lines and transmission substations; distribution substations and feeds to end uses, highest Lifeline weight first; everything else. Ties go to the shortest trip, then the lowest id.
**Registry**
[dynamics.crew_order]



**Drill: How a partial supply passes**
DRAFT wording
B takes power from A at κ = 0.4 and throttles, so B runs at 0.4. C takes telecom from B on a threshold edge and throttles. What is C's state at hour 0?

**Answer**
0: B at 0.4 is below the threshold 0.5, so nothing passes.

**Drill: Throttle, trip and floor**
DRAFT wording
C takes power from A at κ = 0.6 and trips. A is working. What is C's state at hour 0?

**Answer**
0: 0.6 is below the trip point 0.9, so C stops.

**Drill: Weakest supply wins**
DRAFT wording
C takes power from A at κ = 0.6 and water from B at κ = 0.8, and throttles on both. A and B are working. What is C's state at hour 0?

**Answer**
0.6: the lower of 0.6 and 0.8, not their product.

**Drill: Stocks drain and refill**
DRAFT wording
C trips without power and holds 3 h of battery. A, its only power, is damaged at hour 0. At what hour does C go out?

**Answer**
3: the battery covers C for 3 h, then C trips.

**Drill: Telecom fallback**
DRAFT wording
C is a clinic whose telecom comes from A (κ = 1) and throttles. A is damaged at hour 0 and never repaired. What is C's state at hour 30?

**Answer**
0.8: full until hour 24, then $d_{eff} = 1 - 1 \times (1 - 0.8) = 0.8$.

**Drill: Two supplies into one node**
DRAFT wording
C takes power from A and from B, κ = 0.5 each, and throttles. A is damaged. What is C's state at hour 0?

**Answer**
0.5: B's half arrives; A's does not.

**Drill: Crew priority order and ties**
DRAFT wording
One crew. B, a transmission substation 20 km east, and C, a clinic 4 km east, are damaged; each needs 2 h of work. At what hour is C done?

**Answer**
8.9: B first (4.5), then 16 km back to C, 2 h to prepare, 2 h of work.

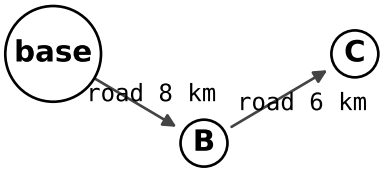
**Drill: The crew sequence**
DRAFT wording
One crew at the base. C, 8 km away by road, is damaged and needs 3 h of work and no parts. At what hour is the work done?

**Answer**
5.2: 0.2 h of travel, 2 h to prepare, 3 h of work.

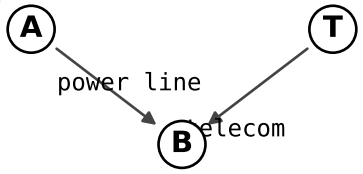
**Drill: Fuel deliveries**
DRAFT wording
C runs on its generator with 6 h of fuel when A, its power, is damaged at hour 0. Depot D is working. At what hour is C back on?

**Answer**
12: the tank runs dry at hour 6; the first delivery lands 12 h after power was lost.

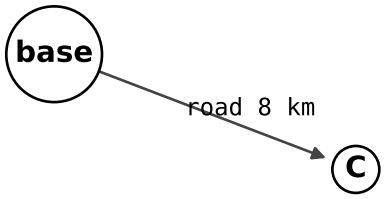
**Travel between jobs**
DRAFT wording
A crew goes from job to job and never back to base. It drives at 40 km/h, walks to a line at 2 km/h, and a sea crossing adds 12 h.
**Registry**
= 40 km/h
  [crews.road_speed_kmh]
= 2 km/h
  [crews.offroad_speed_kmh]
= 12 h
  [crews
    .outlying_island_mobilisation_h]




**Manual switching**
DRAFT wording
A substation that lost all power while its telecom is down cannot be switched back remotely: when power returns, a crew must travel to it and spend 1 h on site.
**Registry**
= 1 h on site
  [repair.manual_switching_h]
= 6 h per failure point
  [storm.line_crew_work_h]




**Time and rounding to steps**
DRAFT wording
Play from event to event at exact times. The engine moves in 15-minute steps: an event takes effect at the first step at or after its exact time. Play ends at hour 72.
**Registry**
= 15 min
  [dynamics.step_min]
= 72 h
  [dynamics.horizon_h]



**Drill: Time and rounding to steps**
DRAFT wording
As in the crew-sequence drill, the crew finishes C's work at hour 5.2. At what hour is C back on in the engine?

**Answer**
5.25: the first 15-minute step at or after 5.2.

**Drill: Manual switching**
DRAFT wording
Line A-B is down 5 km from A (on the road from the base at A), needing 6 h of work. B is a substation 10 km from A; its telecom T is lost for good. One crew at A. At what hour is B back on?

**Answer**
9.5: the line is done at 8.125 and returns at step 8.25; the crew drives 0.125 h to B and switches for 1 h, done at 9.375, back on at step 9.5.

**Drill: Travel between jobs**
DRAFT wording
One crew. B (8 km east) and C (8 km east, 6 km north) are clinics needing 1 h each. At what hour is C done?

**Answer**
6.35: B is nearer, done at 3.2; then 0.15 h to C, 2 h to prepare, 1 h of work.